

Genealogy Dynamics: A Two-Sex Graph-Based Cluster-Dynamics Model of the U.S. Population, 1650–2050

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Adapted from the graph-based cluster-dynamics framework (RadCluster): the pedigree degree is a generation *cluster*, the population is split into paternal and maternal *populations*, and the “reactions” are immigration, cross-population mating, death and emigration.

1 Introduction

This document describes a population-dynamics prototype built by reusing the graph-based modeling approach of RadCluster: the population is partitioned into discrete classes (vertices), and every demographic process is an admissible transition (edge) carrying a rate kernel. The state is integrated as a system of ordinary differential equations from colonial times (**1650**) to the present, and validated against a 1980–2024 hindcast.

In the language of the cluster-dynamics framework, the pedigree degree is a *generation cluster*: each person belongs to a cluster C_d labeled by their pedigree depth $d = 0, \dots, n$, the direct analog of the cluster size in the materials formulation. The population is split into two *populations* — the paternal population P_{pat} and the maternal population P_{mat} — both subsets of the cluster species class C . A cluster state (d, α) is a pedigree-depth- d cluster in population $\alpha \in \{\text{pat}, \text{mat}\}$, with concentration $c_{d,\alpha}(t)$ in millions. The cluster depth n is a free parameter (default $n = 14$, set from the mean generation interval; override with `GENEALOGY_MAX_DEPTH`).

Two alternative definitions of the pedigree depth (child-cluster rules) are carried in parallel:

- **Shallowest inheritance** — the depth counts generations back through which *every* ancestor, on *both* the paternal and maternal sides, was U.S.-born. One immigrant anywhere in the n -generation pedigree breaks the line, so only deep $\times$ deep pairings replenish the deep clusters and C_n is a small endogamous fraction: $g(q, r) = \min(n, 1 + \min(q, r))$.
- **Deepest inheritance** — the child continues the *deeper* of its two parental bloodlines: $g(q, r) = \min(n, 1 + \max(q, r))$. Depth never decreases along a lineage, so mass ratchets upward until it accumulates at the cap C_n .

Two scenario cases are carried: **Case 1**, the 1650–2025 historical run on real demographic data (the featured case), and **Case 2**, a 2025–2050 projection from the Case-1 2025 state with projected inputs (Sec. 7). Three extensions of the base two-sex model are documented: age structure (Sec. 3.4), regional compartments (Sec. 3.5), and a CPS parental-nativity calibration of

α (Sec. 7.8), plus a stochastic colonial-era check (Sec. 7.10) and independent validation against Census data (Sec. 7.11).

2 Related work

Pedigree dynamics has a compact theoretical literature this work builds on. Chang [42] studied a two-parent analog of the Wright–Fisher model and showed the genealogical most recent common ancestor of a population of size n lies $\sim \log_2 n$ generations back, with an “identical ancestors” point at $1.77 \log_2 n$ generations where every surviving lineage becomes an ancestor of everyone — the formal ancestor of the longest-thread argument behind the bloodline rules here. Rohde, Olson and Chang [43] extended this to structured populations with migration by simulation, showing recent common ancestry survives realistic substructure; the regional and religious compartments (Secs. 3.5, 7.9) are the ODE counterpart of their agent model. Derrida, Manrubia and Zanette [44, 45] derived the statistics of pedigree collapse — ancestor repetitions reach a stationary shape after $\sim \log N$ generations — which underlies the cluster-depth wavefront that advances roughly one rung per generation in the colonial runs.

The religious dimension follows the cultural-transmission modeling tradition: Manrubia and Zanette [46] modeled a vertically transmitted cultural trait (the surname) with a birth–death master equation, and Cavalli-Sforza and Feldman [47] founded the quantitative theory of cultural transmission on which group-specific endogamy and disaffiliation transfer rest.

One caveat from that literature is imported explicitly: Matsen and Evans [48] showed genealogical and genetic ancestry run on different timescales, so cluster depth here is a statement about pedigrees, not about genetic contribution.

What is new here is the combination: a forward-time cluster-dynamics ODE framework ported from irradiated-materials modeling, with catalytic mating reactions on an explicit reaction-admissibility graph; two-sex, age, regional, and religious structure; and a 1650–2025 U.S. run grounded in measured demographic series with hindcast validation, plus a 2025–2050 prediction case.

3 Model formulation

3.1 State, clusters and populations

The state vector $\mathbf{c}(t) \in \mathbb{R}_{\geq 0}^{2(n+1)}$ has components $c_{d,\alpha}$ = population of cluster state (d, α) in millions, with cluster (pedigree depth) $d = 0, \dots, n$ and population $\alpha \in \{\text{pat}, \text{mat}\}$ ($n = 14$ by default). The components are block-ordered,

$$\mathbf{c} = [c_{0,\text{pat}}, \dots, c_{n,\text{pat}}, c_{0,\text{mat}}, \dots, c_{n,\text{mat}}]^\top,$$

the paternal (male) block first, then the maternal (female) block. The total population is $N(t) = \sum_{d,\alpha} c_{d,\alpha}(t)$; the cluster totals $T_d(t) = c_{d,\text{pat}}(t) + c_{d,\text{mat}}(t)$ are the quantities directly comparable with the earlier one-population formulation.

3.2 Religious dimension

The U.S. population is further divided into eight religious populations: $P = \{\text{pat}, \text{mat}\} \times G$ with

$$G = \{\text{pro}, \text{cat}, \text{mor}, \text{jew}, \text{mus}, \text{dhr}, \text{oth}, \text{una}\},$$

Protestant, Catholic, Mormon (Latter-day Saint), Jewish, Muslim, Hindu/Buddhist, Orthodox/other faith, and religiously unaffiliated. A cluster state is (d, s, g) with concentration $c_{d,s,g}(t)$ in millions; the state vector $\mathbf{c}(t) \in \mathbb{R}_{\geq 0}^{8 \cdot 2 \cdot (n+1)}$ is block-ordered by group (240 ODEs at $n = 14$).

3.3 Master equation in matrix form

The dynamics take the compact form

$$\frac{d\mathbf{c}}{dt} = \mathbf{P}(t) + \mathbf{S} \mathbf{J}(\mathbf{c}) - \mathbf{D}(t) \mathbf{c}, \quad (1)$$

with the pieces identified as follows.

- **Source vector:** $\mathbf{P}(t) \in \mathbb{R}^{2(n+1)}$ — immigration $I(t)$ (millions/yr) splits into the two populations,

$$P_{(0,\text{pat})} = \sigma(t) I(t), \quad P_{(0,\text{mat})} = (1 - \sigma(t)) I(t),$$

all other entries zero. The immigrant male fraction $\sigma(t)$ is a data-driven series, not a constant: 0.65 *assumed* for 1650–1819; 0.629 *estimated* for 1820–1917 (HSUS period aggregate of 62.9% male, applied annually) [33]; convergence to 0.50 by 1950, *estimated* [33]; 2015–2023 from measured DHS LPR admissions by sex (Table 9) [34] blended with an *estimated* unauthorized-inflow sex proxy (DHS unauthorized-stock sex composition, Table 4) [35]; 0.470 in 2024, *interpolated*. Production runs use the measured $s = 0.512$ and this $\sigma(t)$.

- **Loss matrix:** $\mathbf{D}(t) = \text{diag}(\mu_\alpha(t) + \varepsilon_\alpha)$ — diagonal, with sex-specific mortality $\mu_\alpha(t)$ plus emigration ε_α (uniform by default). The sex *differential* $\mu_{\text{pat}}/\mu_{\text{mat}}$ (2024 ≈ 1.069) is *measured* from HLD sex-specific life tables, 1900–2023 [16]; the *level* is *estimated* (scaled to the aggregate CDR), and pre-1900 uses the 1900–1919 mean ratio.
- **Flux vector:** $\mathbf{J}(\mathbf{c}) \in \mathbb{R}^{(n+1)^2}$, indexed by paternal/maternal parent pairs (q, r) ,

$$J_{qr}(\mathbf{c}) = B(\mathbf{c}) \left[\alpha \delta_{qr} m_q^{\text{pat}}(\mathbf{c}) + (1 - \alpha) m_q^{\text{pat}}(\mathbf{c}) m_r^{\text{mat}}(\mathbf{c}) \right], \quad (2)$$

where the mating weights are

$$m_q^{\text{pat}}(\mathbf{c}) = \frac{\beta_q^{\text{pat}} c_{q,\text{pat}}}{\sum_p \beta_p^{\text{pat}} c_{p,\text{pat}}}, \quad m_r^{\text{mat}}(\mathbf{c}) = \frac{\beta_r^{\text{mat}} c_{r,\text{mat}}}{\sum_p \beta_p^{\text{mat}} c_{p,\text{mat}}},$$

(a zero denominator returns zero weights) and the total births are marriage-function style,

$$B(\mathbf{c}) = \min \left(\sum_p \beta_p^{\text{pat}} c_{p,\text{pat}}, \sum_p \beta_p^{\text{mat}} c_{p,\text{mat}} \right),$$

With probability α the father's cluster equals the mother's (assortativity); otherwise the parents are drawn independently.

- **Stoichiometry matrix:** $\mathbf{S} \in \mathbb{R}^{2(n+1) \times (n+1)^2}$ with

$$S_{(g(q,r),\text{pat}), (q,r)} = s, \quad S_{(g(q,r),\text{mat}), (q,r)} = 1 - s, \quad (3)$$

all other entries zero, where $g(q, r)$ is the child-cluster rule and $s = 0.512$ is the *measured* NCHS sex ratio at birth (≈ 1.05 male per female [4]). Each column of $\mathbf{S}$ sums to $s + (1 - s) = 1$, so the mating flux conserves people: $\sum_{d,\alpha} (S J)_{d,\alpha} = B$ exactly. The child-cluster rules are

$$\text{shallowest inheritance: } g(q, r) = \min(n, 1 + \min(q, r)), \quad (4)$$

$$\text{deepest inheritance: } g(q, r) = \min(n, 1 + \max(q, r)). \quad (5)$$

The parents act catalytically (not consumed), so each S column has exactly two nonzero entries; S is rule-dependent but time-independent, while $J(c)$ carries the nonlinearity. The reaction-kernel table (Sec. 4) is S written column-wise. Summing (1) over d and α gives the exact balance

$$\frac{d}{dt} \sum_{d,\alpha} c_{d,\alpha} = I + B - \sum_{\alpha} (\mu_{\alpha} + \varepsilon_{\alpha}) \sum_d c_{d,\alpha},$$

reducing to $dN/dt = I + B - (\mu + \varepsilon)N$ for uniform sinks. The time-dependent inputs $I(t)$, $\beta(t)$ and $\mu(t)$ are compiled from public demographic sources and cached with per-row quality flags [1, 2, 3, 5, 6, 8].

Symmetry reduction. With $s = \sigma = 0.5$, equal paternal and maternal fertility, uniform sinks, and a symmetric initial condition, the two populations stay identical: B reduces to the old one-sex form and the aggregated totals $T_d = c_{d,\text{pat}} + c_{d,\text{mat}}$ satisfy exactly the one-population equations. A side-by-side integration of the two-sex right-hand side against the pre-refactor one-sex implementation agrees to 4.3×10^{-9} (millions), so the refactor is a shallowest generalization. Production runs use the measured $s = 0.512$.

Religious mating. With the religious dimension the flux is indexed by parent groups as well as clusters, $J_{(q,g_1),(r,g_2)}(c)$, with

$$J_{(q,g_1),(r,g_2)} = B P(g_1, g_2) \left[\alpha \delta_{qr} w_{q|g_1}^M + (1 - \alpha) w_{q|g_1}^M w_{r|g_2}^F \right], \quad (6)$$

where $A_{q,g}^M = \beta_q c_{q,\text{pat},g}$ is the marriageable mass of paternal cluster q in group g (analogously $A_{r,g}^F$), $A_g^M = \sum_q A_{q,g}^M$, $A^M = \sum_g A_g^M$, $B = \min(A^M, A^F)$, and $w_{q|g_1}^M = A_{q,g_1}^M / A_{g_1}^M$ is the within-group cluster weight (zero denominators guarded to zero). The religious pairing

$$P(g_1, g_2) = \pi_{g_1}^M \left[\rho_{g_1} \delta_{g_1 g_2} + (1 - \rho_{g_1}) \pi_{g_2}^F \right], \quad (7)$$

with $\pi_{g_1}^M = A_{g_1}^M / A^M$ and $\pi_{g_2}^F = A_{g_2}^F / A^F$: the father's group is drawn first, then with group-specific endogamy ρ_{g_1} the mother is either kept in the same group or drawn from the female availability shares. The bracket in (6) is the usual class-assortativity kernel within each group–sex; fertility is uniform across groups (group fertility differentials are not modeled). Summing (6) over all indices gives B exactly, so the flux is conservative. The stoichiometry sends the $s:(1-s)$ split of each (q, g_1, r, g_2) pairing's births to the child cluster $g(q, r)$ in the *mother's* group g_2 — maternal religious transmission is an *assumption*. With $\rho_g = 0$ the group-aggregated right-hand side coincides with the two-sex right-hand side to 4.2×10^{-17} (millions): a shallowest generalization; any $\rho_g > 0$ genuinely reshapes the aggregate pedigree dynamics through religious assortment.

Disaffiliation. A further RAG edge class transfers $(d, s, g) \rightarrow (d, s, \text{una})$ for $g \in \{\text{pro, cat, mor, oth}\}$ at rate $\delta(t)$: zero before 1970, ramping to δ_{max} over 1970–1990, constant thereafter (internal; conserves people to $\sim 10^{-15}$). $\delta_{\text{max}} = 8 \times 10^{-3} \text{ yr}^{-1}$ is *calibrated* to the Pew 2023–24 unaffiliated share ($\approx 29\%$) [9].

3.4 Age structure

The state is $c_{d,s,a}(t)$: cluster $d = 0, \dots, 14 \times \text{sex } s \times 5\text{-year age band } a = 0, \dots, 17$ (0–4 through 85+): $15 \times 2 \times 18 = 540$ states. The master equation (1) gains one edge class: aging transfers $\gamma_{a-1} c_{d,s,a-1}$ with $\gamma_a = 1/5 \text{ yr}^{-1}$ (continuous Leslie-style transfer; the exponential sojourn's CV =

1 vs ≈ 0 for a true 5-year band is a flagged limitation). Births enter only $a = 0$, from bands 3–9 (ages 15–49); the maternal mating weight is $w^{\text{mat}}[q] = \sum_a \beta_a(t) c_{q,\text{mat},a}$ with the age-specific fertility rate $\beta_a(t)$, and the paternal weight uses the female ASFR shape as an *assumed* proxy (no male ASFR series exists). The pair feeds the unchanged aggregate mating flux and the verbatim child rules, so the pedigree rules are preserved exactly. Mortality is sex- and age-specific, $\mu_{s,a}(t)$; emigration ε is uniform; immigration enters C_0 with the $\sigma(t)$ series of Sec. 2.1 and an *assumed* fixed age profile (peak 20–39). The exact balance $dN/dt = I + B - D - \varepsilon N$ holds to 10^{-10} on random states, and age-uniform rates reproduce the aggregate cluster totals to 6×10^{-9} M, so the extension is a shallowest generalization.

Provenance (every input labeled; never present an assumption as an estimate). Fertility $\beta_a(t)$: UN World Population Prospects 2024 ASFR, USA 1950–2023, *estimated* [15]; 2024 held, *interpolated*; 1800–1949 Haines TFR anchors (1800: 7.04) $\times$ fixed 1950 age pattern, *interpolated* $\times$ *assumed*; 1650–1799 TFR 7.04 held $\times$ 1950 shape, *assumed* [5]. Mortality $\mu_{s,a}(t)$: Human Life-Table Database (HLD) USA $m(x)$ 1900–2023, *measured* [16] — the HLD tables compile Bell & Miller SSA Actuarial Study 116 (1900–2000) [17] and NCHS/NVSR life tables (2001–) [18]; 2024 held, *interpolated*; 1650–1899 level *calibrated* to Haines CDR targets (age pattern *assumed*). Sex ratio at birth $s = 0.512$, *measured* (NCHS) [4]; immigrant male fraction $\sigma(t)$ per the series of Sec. 2.1 (not 0.5); ε *calibrated* as in the aggregate model. Spot diagnostics: TFR 3.11 (1950), 3.71 (1957), 1.79 (1976), 2.09 (2007), 1.62 (2023); mean age at childbearing 26.7 yr (1950), 26.0 (1980), 29.9 (2023).

3.5 Regional compartments

The state is $c_{d,s,r}$: $15 \times 2 \times 4 = 120$ states over the four Census regions (Northeast, Midwest, South, West). Mating is entirely *within* region (region assortativity = 1, *assumed*; cross-region unions are future work) using the unchanged kernel; immigration splits by region, $I_r(t) = I(t) \lambda_r(t)$, into C_0 ; and a conservative transfer operator moves people between regions,

$$(Tc)_{d,s,r} = \sum_{r' \neq r} m_{r' \rightarrow r} c_{d,s,r'} - \sum_{r' \neq r} m_{r \rightarrow r'} c_{d,s,r},$$

with the same rates for all clusters and sexes (no cluster-specific flow data). With migration off, the regional model aggregates *exactly* to the national one (max cluster deviation 6.4×10^{-14} M); with migration on, the total-balance identity holds to 1.3×10^{-15} .

Provenance. Inter-regional migration rates from the Census ACS 1-year state-to-state migration flows, 2023 and 2024, *measured* [19]: gross inter-regional flow 4.06 and 3.75 M/yr (largest corridors point South: West→South 0.67 M/yr in 2023); pre-2023 held at 2023, *assumed*. Immigration regional allocation $\lambda_r(t)$ from DHS Yearbook 2023 Table 4 (LPR state of residence, FY2014–2023), *measured* [20] — applied to all of $I(t)$ including unauthorized, *assumed* (2014 shares: NE 0.263, MW 0.118, S 0.336, W 0.283; 2023: 0.240, 0.113, 0.379, 0.268). The 1650 IC (55% NE / 45% S) is an *assumed* spin-up.

4 Population-dynamics graphs

The reaction graphs below are generated *from the model code* (NetworkX, fixed seed 20260926), not hand-drawn: every vertex is a real state variable, every edge a real term of the right-hand side. Rectangles are paternal states, ellipses maternal; diamonds are mating-pair vertices M_{qr} ; child-edge colors encode rule agreement (shallowest-only, deepest-only, both). Full legends are drawn on each figure: Fig. 1 the two-sex base graph ($n = 4$ shown, 26 states); Fig. 2 the reli-

gious extension ($n = 2$ shown, 240 ODEs at $n = 14$); Fig. 3 the age-structured model ($n = 2$, ages 0–4 to 20–24 shown; 540 states); Fig. 4 the regional model ($n = 2$ shown; 120 states).

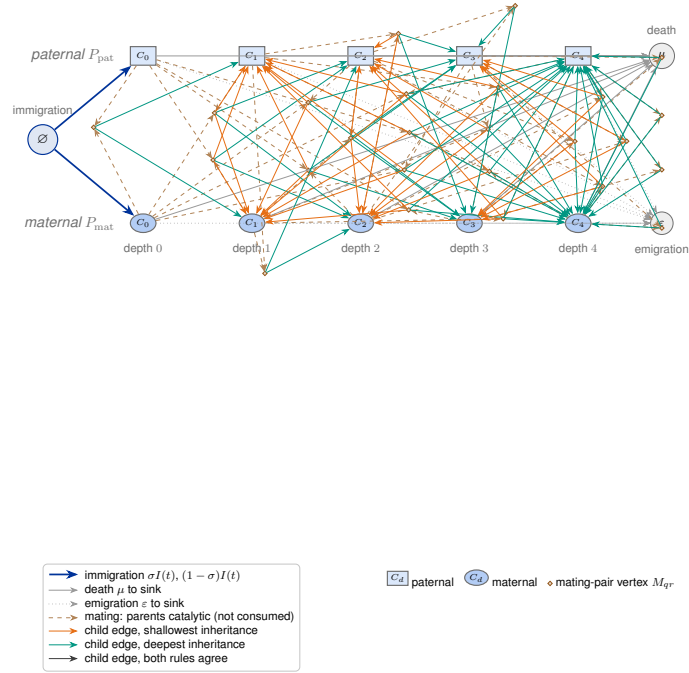


Figure 1: Reaction graph of the two-sex model (representative $n = 4$), generated from `genealogy` code. Vertices: cluster states (d, α) , immigration source, death/emigration sinks, mating-pair diamonds M_{qr} . Edges: immigration into C_0 , catalytic mating, child edges colored by rule agreement, death and emigration to sinks.

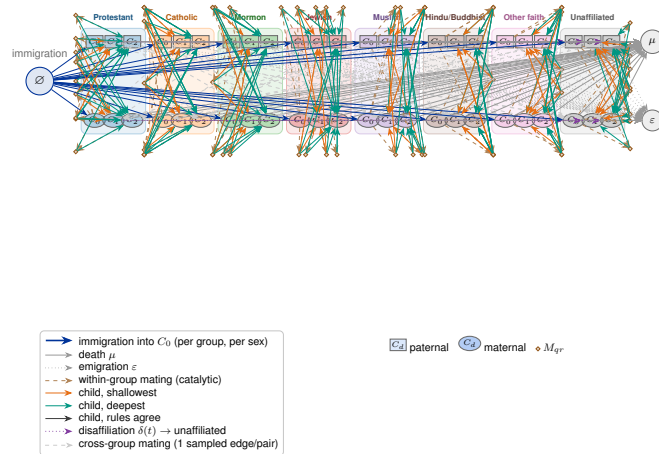


Figure 2: Reaction graph of the religious extension ($n = 2$ shown; 240 ODEs at $n = 14$), generated from `genealogy` code. Adds group-colored state lanes, within-group mating, sampled cross-group mating edges, and the disaffiliation transfers $(d, s, g) \rightarrow (d, s, \text{una})$ at rate $\delta(t)$.

5 Reaction kernel table

Every admissible transition (edge class) of the two-lane graph, with its stoichiometry and rate kernel — the direct analog of the RadCluster reaction table. Parents are catalytic; the stoichiometry matrix S (one column per (q, r) pairing) carries the $s : (1 - s)$ birth split.

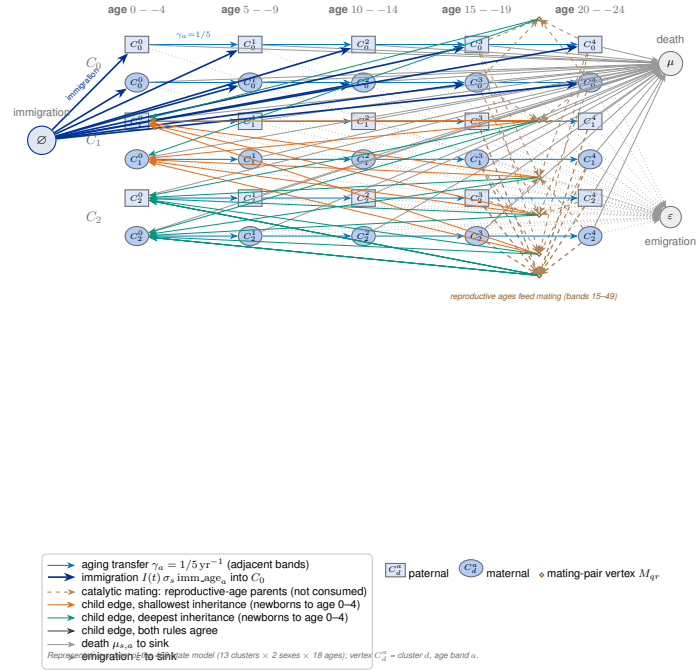


Figure 3: Reaction graph of the age-structured model (representative subset: $n = 2$, ages 0–4 to 20–24 shown; the full model has 540 states), generated from `genealogy.age` code. New edge class: aging transfers $(d, s, a) \rightarrow (d, s, a + 1)$ at $\gamma_a = 1/5/\text{yr}$; reproductive-age parents feed the mating diamonds; newborns enter age band 0.

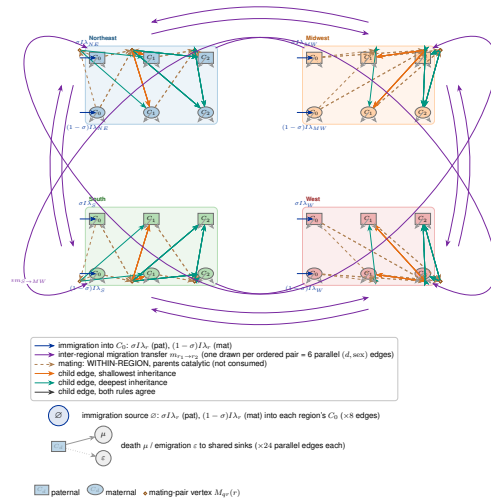


Figure 4: Reaction graph of the regional model (representative $n = 2$; the full model has 120 states), generated from `genealogy.regional` code. Mating is within-region (catalytic); each ordered region pair carries one drawn transfer arrow $m_{r_1 \rightarrow r_2}$ (six parallel (d, sex) edges in the model); immigration splits by region into C_0^g .

#	Process class	Transition	Kernel / flux
1	Immigration (source)	$\emptyset \rightarrow (0, \alpha)$	$\sigma_\alpha I(t)$, $I = I_{\text{LPR}} + I_{\text{unauth}}$ [M/yr]
2	Birth (mating)	$(q, \text{pat}) + (r, \text{mat}) \rightarrow s(g, \text{pat}) + (1-s)(g, \text{mat})$	J_{qr} of Eq. (2), $g = g(q, r)$ (parents catalytic)
3	Mortality (sink)	$(d, \alpha) \rightarrow \emptyset$	$\mu_\alpha(t) c_{d,\alpha}$
4	Emigration (sink)	$(d, \alpha) \rightarrow \emptyset$	$\varepsilon_\alpha c_{d,\alpha}$

Child-cluster kernel $g(q, r)$: shallowest inheritance $g = \min(1 + \min(q, r), n)$ (both parents must be deep); deepest inheritance $g = \min(1 + \max(q, r), n)$ (the child takes the deeper parental bloodline). Sex ratio at birth $s = 0.512$ is measured from NCHS natality data [4]; the immigrant male fraction $\sigma(t)$ is the data-driven series of Sec. 2.1 (DHS LPR by sex [34] for 2015–2023, HSUS 1820–1917 [33], assumed colonial).

6 Input parameters and assumptions

Parameter	Symbol	Value / series
LPR immigration	$I_{\text{LPR}}(t)$	FY1980–FY2024
Unauthorized inflow	$I_{\text{unauth}}(t)$	1980–2024
Colonial immigration	$I(t)$	1650–1819
Fertility	$\beta^\alpha(t)$	2 CBR(t)/1000, same schedule both sexes
Mortality	$\mu_\alpha(t)$	CDR(t)/1000 scaled by the <i>measured</i> HLD sex differential (2024 $\mu_{\text{pat}}/\mu_{\text{mat}} \approx 1.069$)
Emigration	ε_α	$7.5 \times 10^{-4} \text{ yr}^{-1}$, uniform across sexes (default)
Sex ratio at birth	s	0.512 (≈ 1.05 male per female)
Immigrant M fraction	$\sigma(t)$	0.65 (1650–1819); 0.629 (1820–1917); $\rightarrow 0.50$ by 1950; DHS LPR by sex + unauth. proxy 2015–
Assortativity	α	0.8 enforced central; swept 0.0, 0.8, 0.9; CPS-calibrated 0.66
Age fertility	$\beta_a(t)$	18 bands, nonzero ages 15–49
Age mortality	$\mu_{s,a}(t)$	sex $\times$ age, 1900–2023
Inter-regional migr.	$m_{r \rightarrow r'}$	12 ordered pairs, 2023–24
Immigr. allocation	$\lambda_r(t)$	4 regions, FY2014–23
Case-2 β, μ	trend	CBR 10.26 $\rightarrow$ 5.26; CDR 9.20 $\rightarrow$ 11.09/1000
Case-2 I, δ	frozen/held	$I = 1.705 \text{ M/yr}$; $\delta = 8 \times 10^{-3} \text{ /yr}$
Religious endogamy	ρ_g	pro .81, cat .75, mor .87, jew .62, mus .84, una .68; dhr .85, oth .80
Immigrant religion	$\iota_g(t)$	era brackets: colonial pro .93; 1800–60 pro .68/cat .27; 1860–1920 pro .48/cat .35/jew .10; 1920–
Disaffiliation	$\delta(t)$	0 before 1970; ramp 1970–90 to $\delta_{\text{max}} = 8 \times 10^{-3} \text{ yr}^{-1}$

Parameter	Symbol	Value / series
Cluster depth	n	14 (env. GENEALOGY_MAX_DEPTH)
Initial condition	$\mathbf{c}(1650)$	$C_0=50,368$ split $\sigma/(1-\sigma)$ M/F, rest 0
Population anchors	$N(t)$	1650–1790 HSUS; decennial 1790–2020

Validation: the 1980–2024 hindcast reproduces the U.S. total population within **2.87 M maximum error** (2024: -2.87 M, -0.8%), after adding unauthorized inflows and calibrating ε ; the two rules’ totals agree to 7×10^{-4} M, i.e. totals are essentially rule-independent [7, 3, 8]. The 1650–2025 colonial run’s anchor fit is discussed in Sec. 6; 1650–1750 is a spin-up transient. Pre-1800 vital rates are assumptions, flagged per row in `data/inputs.csv`.

7 Results

Scenario structure. **Case 1 (historical, real data):** the 1650–2025 colonial run on the measured pipeline of Sec. 5 — the featured case. **Case 2 (projection):** 2025→2050 from the exact Case-1 2025 handoff state with projected inputs: $\beta(t)$ by OLS trend on the measured 2015–24 crude birth rate (*projected*; CBR 10.26 → 5.26/1000 in 2050), $\mu(t)$ by OLS trend on the measured 2015–19 and 2023–24 crude death rates, excluding the COVID years 2020–22 (*projected*; CDR 9.20 → 11.09/1000), $I(t)$ frozen at the 2024 measured 1.705 M/yr (*assumed*), $\varepsilon = 7.5 \times 10^{-4}$ /yr (*calibrated*), $\sigma(t)$ held at the 2024 blend 0.4703 (*projected*), $\mu(t)$ projected by OLS with the 2024 sex differential held, and (religious) δ held at the calibrated 8×10^{-3} /yr (*projected* continuation; one-way caveat). Rate continuity at the handoff is smooth: projected CBR 10.26 vs 10.6 measured, CDR 9.20 vs 9.0, I exact. **2050 totals: 279.7 M in every run** (+8.0% over the 2025 258.95 M) — rule- and α -independent, since the mating flux conserves people.

Deepest-cluster shares in 2050 ($n = 14$; 2025 Case-1 values in parentheses):

Model	Rule	$\alpha=0.0$ (null)	0.8 (central)	0.9 (bound)
two-sex, $n=14$	shallowest	0.00% (0.00%)	0.42% (0.45%)	1.34%
	deepest	65.29% (71.88%)	16.66% (18.35%)	8.61% (9.44%)
religious, $n=14$	shallowest	0.00%	0.42% (0.45%)	1.35%
	deepest	62.31% (68.80%)	16.25% (17.93%)	8.52% (9.34%)
regional, $n=14$	shallowest	0.00% (0.00%)	0.43% (0.47%)	1.38%
	deepest	65.07% (71.65%)	16.76% (18.47%)	8.73% (9.57%)

Deep shares ease slightly 2025→2050: the frozen immigration (1.705 M/yr, all into C_0) outpaces deep-cluster replenishment under the declining-fertility trend (Fig. 10).

Assortativity bracket. $\alpha = 0.0$ is the random-mating null; $\alpha = 0.8$ the enforced central value, anchored to the mean religious endogamy $\bar{\rho}_g \approx 0.78$; $\alpha = 0.9$ a strong-endogamy bound. Under both rules deep-cluster shares rise with α (Fig. 10), because only deep×deep pairings replenish C_n . The deepest rule is shallowest-like in direction (rising with α) but sits well above the shallowest values: forgiving one foreign-born parent per generation lifts the deep share. CPS parental-nativity data calibrate $\alpha_{\text{CPS}} = 0.66$ (Sec. 7.8); $|\cdot|$ is not material, so the enforced 0.8 is kept as central — now bracketed by data (ratio fit 0.66; absolute-series preference ≥ 0.8).

7.1 Time-varying assortativity $\alpha(t)$: history dominates the endpoint

A time-varying assortativity schedule was built from the marriage/intermarriage literature as a *documented sensitivity variant* (not the default). The proxy mapping is approximate: pedigree endogamy $\approx 1 - k x(t)$, where $x(t)$ is the measured Pew newlywed interracial/interethnic share and the slope $k = 1.8462$ is *calibrated* to reproduce $\alpha_{\text{CPS}} = 0.66$ at 2013. The schedule (Fig. 6, left) runs 0.95–0.94 in 1650–1776 (*assumed*, no marriage data); 0.93 (1850) and 0.94 (1910) *estimated* from 19th-century ethnic and immigrant endogamy [30, 27, 28, 32, 29]; 0.945 (1940) *estimated* [31]; 0.956→0.686 over 1960–2015 from the measured Pew newlywed intermarriage series (2.4%→17%) [24, 25], corroborated (not calibrated) by the religious intermarriage series [26]; 0.653 in 2025 *estimated*; 0.653 *assumed* Case-2 hold through 2050. Re-running the Case-1 colonial model at $n = 14$ with $\alpha(t)$ gives 2025 C_{14} of **1.87%** shallowest / **7.78%** deepest, versus 0.45% / 18.35% at constant $\alpha = 0.8$ (Fig. 6). *Path dependence*: $\alpha(t)$ sits at 0.93–0.96 for three centuries, so deep clusters compound mass long before the post-1960 decline begins; the decline cannot undo the accumulated stock. Under the deepest rule the schedule *undershoots* constant-0.8 because $\alpha(t)$ ends at 0.653, below the central value, when recent mixing matters most.

Recommendation: keep $\alpha = 0.8$ as the default. The schedule stays opt-in as a sensitivity: it quantifies how wrong constant- α could be if historical endogamy was as strong as the literature suggests. Promoting $\alpha(t)$ would let colonial anchors drive the headline numbers, an assumption dressed as a result.

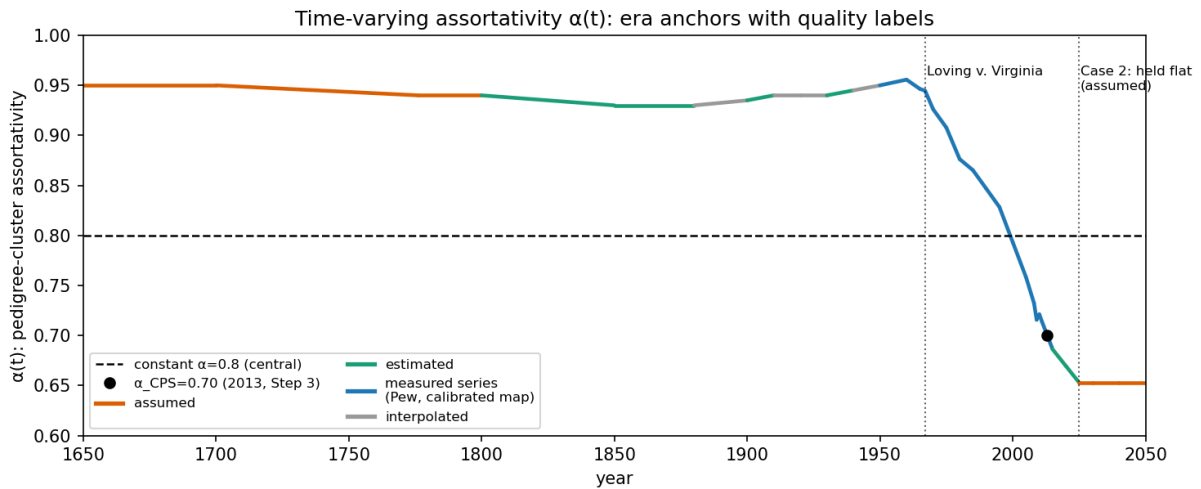


Figure 5: The $\alpha(t)$ schedule 1650–2050 with quality-colored segments, constant-0.8 reference, $\alpha_{\text{CPS}} = 0.66$ (2013) marker, and the Case-2 hold.

Maternal bottleneck. The $\min(W^{\text{pat}}, W^{\text{mat}})$ marriage function makes the *maternal* population the limiting sex under male-skewed immigration, sharply reducing births vs the old $\sigma = 0.5$ assumption: decomposition at shallowest $\alpha = 0.8$ gives 2025 totals 300.2M (old σ , old μ), 241.5M (new σ , old μ), 324.9M (old σ , new μ), and **259.0M** (new σ , new μ) — all four model families now land at ≈ 259 M in 2025. The anchor fit is worse than the old run: worst error -29.3% **at 1830** (old: -13.4%). The emigration rate ε is deliberately *not* recalibrated to close the gap — the mismatch is reported honestly rather than tuned away. The hindcast, which starts from the measured 1980 state, is barely affected (-2.87 M, -0.8% in 2024).

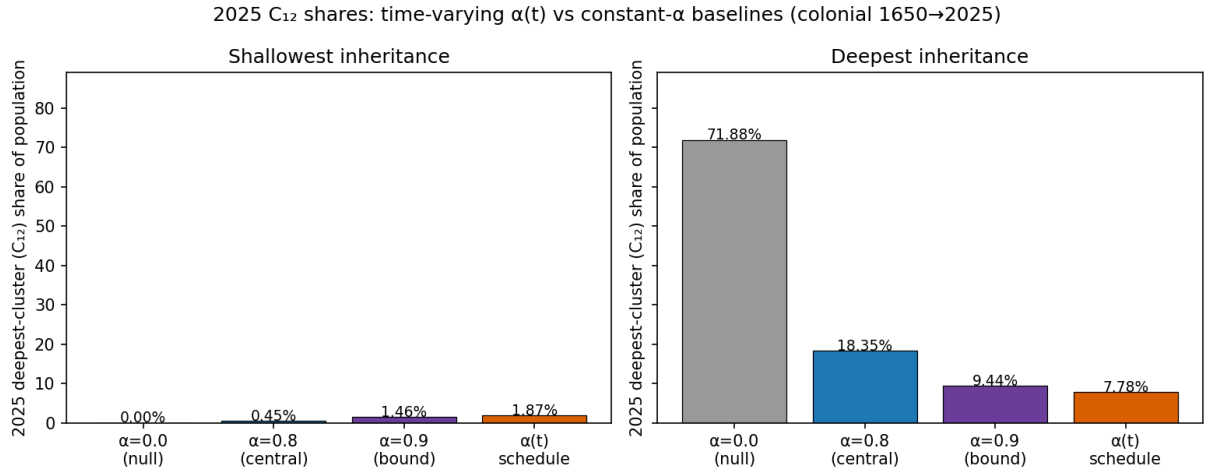


Figure 6: 2025 C_{14} shares — constant- α bracket (0.0 null, **0.8** central, 0.9 bound) vs the $\alpha(t)$ schedule, $n = 14$. Shallowest: the schedule exceeds the 0.9 bound (path dependence). Deepest: the schedule undershoots constant-0.8 (recent mixing dominates).

7.2 Sex-specific rates: the immigrant male-share series

The old $\sigma = 0.5$ assumption is replaced by the data-driven $\sigma(t)$ series of Fig. 7 (construction in Sec. 2.1): 0.65 *assumed* (1650–1819, indentured-labor skew), 0.629 *estimated* (1820–1917, HSUS period aggregate applied annually), quota-era convergence to 0.50 by 1950 *estimated*, 2015–2023 measured DHS LPR-by-sex [34] blended with an *estimated* unauthorized-stock-as-inflow proxy [35], 2024 0.470 *interpolated*, held through 2050 *assumed*. Sex mortality differential $\mu_{\text{pat}}/\mu_{\text{mat}}$ (2024 ≈ 1.069) is *measured* [16]; the level is scaled to the aggregate CDR (*estimated*). All model families now converge on a 2025 total of ≈ 259 M (two-sex 258.95, religious 258.95, regional 258.95, age 259.7 M). The largest remaining sex-structure uncertainty is the paternal age schedule (still the female ASFR shape, *assumed*): shifting it ~ 2 yr older moves 2025 C_n by 2–5 pp at $\alpha \geq 0.8$ (Sec. 6.4), because the $\min(W^{\text{pat}}, W^{\text{mat}})$ kernel makes the older-shifted paternal side the binding sex more often.

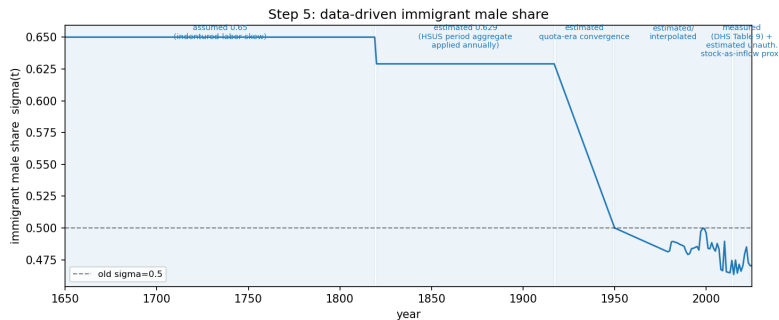


Figure 7: The immigrant male-share series $\sigma(t)$ used by all families (labels in Sec. 2.1). Two era breaks are visible: 1820 (the HSUS 0.629 aggregate takes over from the assumed 0.65) and 1950 (parity reached), then the DHS measured + proxy era from 2015 (0.470 in 2024).

7.3 Case 1: colonial run, 1650–2025

The historical run (Figs. 8, 9) gives the 2025 ordering at $n = 14$: shallowest C_{14} is 0.00%, **0.45%**, 1.46% for $\alpha = 0.0, 0.8, 0.9$; deepest 71.88%, **18.35%**, 9.44%. The two-sex and religious $n = 14$ models agree to 0.004 pp at central α (shallowest), making the four model families apples-to-apples. The paternal share of the 2025 population is 50.09% in every run.

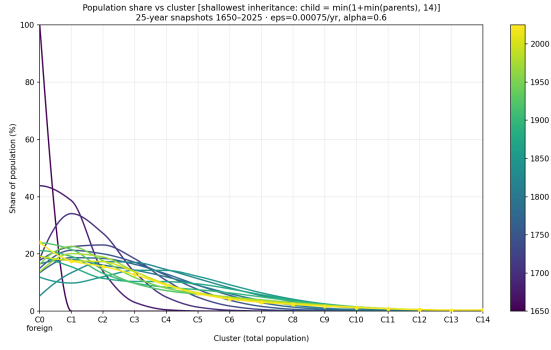


Figure 8: (a) Shallowest: C_{14} reaches **0.45%** by 2025 ($\alpha = 0.8$); deep clusters replenish only through deep $\times$ deep intermarriage.

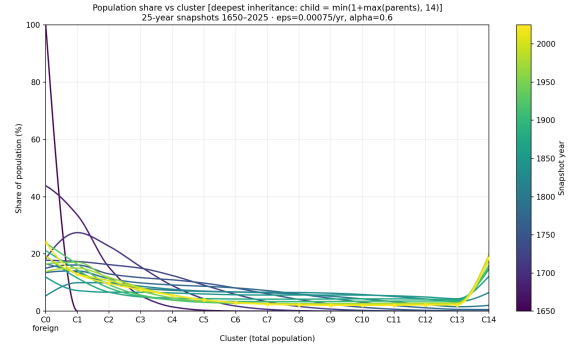


Figure 9: (b) Deepest: C_{14} reaches **18.35%** by 2025; with no downward pull, lineages ratchet upward and accumulate at the cap.

7.4 Case 2: 2025–2050 projection

From the exact Case-1 2025 handoff state: the 2050 deepest-cluster bar chart by rule and α (Fig. 10) and the 2050 totals/deepest shares table above. The religious Case-2 projection is reported with the religious results (Sec. 7.9).

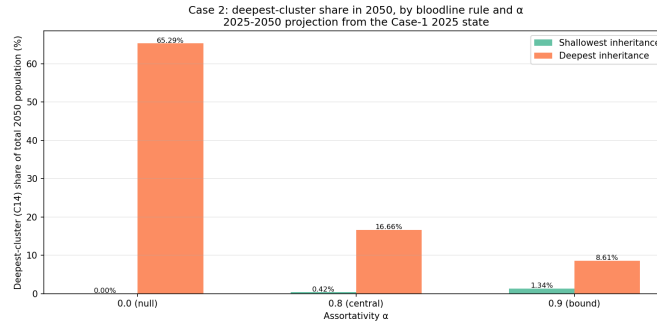


Figure 10: Deepest-cluster (C_{14}) share of the 2050 population vs assortativity α , by bloodline rule (Case-2 projection 2025–2050 from the Case-1 2025 state). Bracket: 0.0 (null), **0.8** (central), 0.9 (bound).

7.5 Deeper cluster resolution: $n = 14$ and the resolved tail

The cluster depth is now $n = 14$ in *all* model families — the religious runs no longer differ from the rest (Sec. 6.8’s old $n = 12$ vs $n = 14$ incomparability is gone). $375/14 = 26.8$ yr per generation matches the measured mean generation interval *independently*: the truncation choice is self-consistent, not just convenient. Four runs of the two-sex Case-1 model at $\alpha = 0.8$ (Fig. 11):

Rule	$n=12$	$n=14$	$n=18$	$n=24$
Shallowest C_n	1.33%	0.45%	0.03%	0.00%
Deepest C_n	22.74%	18.35%	9.33%	1.31%

Totals stay bit-identical across n (mass is only relabeled), and the cap is *bit-conserved*: C_{14} at $n = 14$ equals $\sum_{d \geq 14}$ at $n = 18$ and $n = 24$ under *both* rules (0.4488% / 18.3527%) — no mass piles up at the truncation. Under shallowest inheritance the tail decays geometrically: beyond C_{18} only 0.034% remains. Under deepest inheritance the cap *intentionally* accumulates the ratchet mass — this is the rule’s physics, not a numerical artifact — and resolving it to $n = 24$ shows the deep tail is nearly flat ($\sim 2.2\%$ per cluster from C_{12} to C_{17} , declining to 0.68% at C_{23}); the invariant is $\sum_{d \geq 14}$, which is conserved exactly. **Verdict:** $n = 14$ is a faithful, bit-conserved

truncation under both rules; the cap does not distort the dynamics, and the continuous-index alternative is unnecessary.

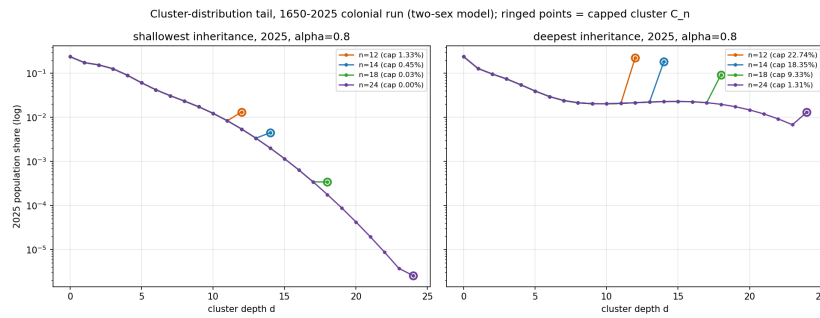


Figure 11: Tail of the 2025 cluster distribution at $\alpha = 0.8$, $n = 12/14/18/24$ (ringed = capped cluster). The cap is bit-conserved: C_{14} at $n = 14$ equals $\sum_{d \geq 14}$ at $n = 18$ and $n = 24$ — no pile-up. At $n = 24$ the tail beyond C_{18} is exhausted.

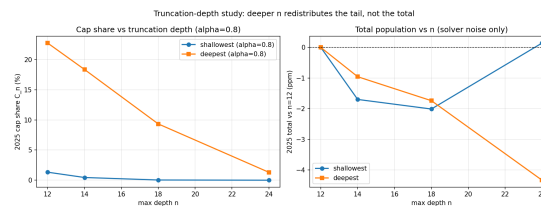


Figure 12: 2025 C_n share at $\alpha = 0.8$ for $n = 12, 14, 18, 24$, both rules. $n = 14$ is the operating point; $n = 12$ inflated the deep share $\sim 3\times$ by truncating the tail.

7.6 Age structure: resolved tail amplifies deep concentration

The 540-state colonial run at $\alpha = 0.8$ (Sec. 3.4; $n = 14$): 2025 C_{14} is **1.62%** (age) vs 0.45% (aggregate) under the shallowest rule and **38.90%** vs 18.35% under the deepest rule (Fig. 14) — age structure concentrates deep mass in old cohorts, and the deepest rule ratchets that reservoir to the cap. The wavefront experiment (Fig. 13; closed population, constant rates) remains valid as a *rate* measurement, independent of n : the age front advances $\sim$ one rung per human generation (floor at the minimum parental age, 15 yr); under high colonial fertility the aggregate climbs $\approx 1.9\times$ faster (no floor), while under low modern fertility it is mass-accumulation-limited and runs slower. 2025 totals: age-structured 259.7 M vs aggregate 258.95 M vs ~ 342 M actual; anchor validation worst error -24.0% in 2024.

Male-fertility sensitivity (age-structured model). The default equal-schedule proxy (female ASFR shape for fathers, *assumed*) vs a paternal schedule shifted ~ 2 yr older: shallowest $\alpha = 0.8$ 2025 C_{14} 1.62% $\rightarrow$ 0.79% (-0.83 pp, -51% rel.), deepest 38.90% $\rightarrow$ 28.66% (-10.23 pp, -26% rel.); totals -4.9% everywhere (older-shifted paternal weights make the paternal side the binding sex in the min kernel more often). A material modeling uncertainty (≈ 1 – 10 pp on 2025 C_{14} at $\alpha \geq 0.8$): the proxy is retained as default but stays labeled *assumed*.

7.7 Regional compartments: the Midwest runs deepest

Case-1 2025 C_{14} by region at $\alpha = 0.8$ (Sec. 3.5; national 0.47% / 1.40%):

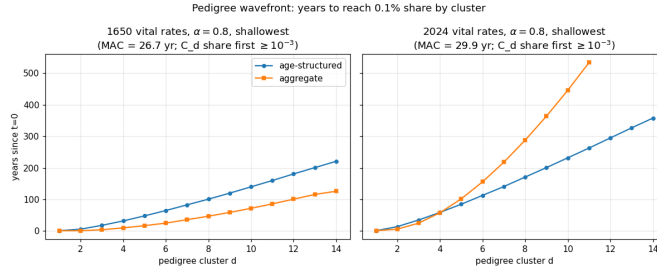


Figure 13: Pedigree wavefront: years for each cluster to reach 0.1% share (closed population, constant rates, shallowest rule, $\alpha = 0.8$). Left (1650 rates): the aggregate front climbs $\approx 1.9\times$ faster — the missing generation-delay artifact (no 15 yr parental-age floor). Right (2024 rates): the aggregate is mass-accumulation-limited and runs slower.

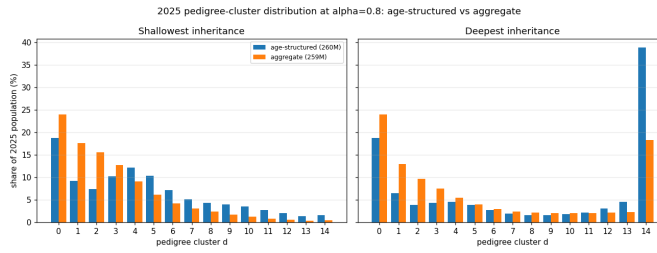


Figure 14: 2025 cluster distributions at $\alpha = 0.8$, $n = 14$: age-structured vs aggregate, shallowest (left) and deepest (right). C_{14} : 1.62% vs 0.45% (shallowest), 38.90% vs 18.35% (deepest) — age structure amplifies deep concentration, especially under the deepest rule.

Rule	NE	MW	South	West
Shallowest	0.36%	0.53%	0.49%	0.43%
Deepest	1.14%	1.54%	1.47%	1.31%

The ordering $MW > South > West > NE$ holds at all three α and both rules (Figs. 15, 16): the Midwest receives the smallest immigration share (11.3%) so its deep lineages are least diluted, the Northeast (24.0%) most diluted; all regions were seeded from the NE/South founder pool via the measured migration network. Regions track together until ~ 1800 , then diverge; all peak ~ 1980 and decline under immigration dilution. Region totals are emergent, not calibrated (2025: NE 39.3, MW 45.5, South 115.9, West 58.2 M) — regional vital-rate differences and calibration are future work. Case-2 2050: ordering unchanged (shallowest NE 0.34 / MW 0.49 / S 0.45 / W 0.40%, national 0.43%; deepest 13.37 / 18.93 / 17.67 / 15.62%, national 16.76%).

7.8 CPS parental-nativity calibration of α

CPS ASEC 2024 Table 1 gives the annual second-generation share 2005–2024, 10.5%→12.6% (*measured* [21]); P23-214 gives the 2013 split: second generation 11.7%, of whom 59% have two foreign-born parents, i.e. 6.90% of the population (*measured* × *measured* [22]). Model mapping: shallowest C_1 = US-born with ≥ 1 foreign-born parent (the CPS second generation); deepest C_1 = US-born children of *two* immigrants. The calibration statistic is the ratio deepest- C_1 /shallowest- C_1 versus the CPS 59%: the underlying people are identical in both runs (only the cluster *labeling* differs by rule), so the ratio is the model’s exact analog of the CPS split and cancels the colonial run’s total-population level error. Result: $\alpha_{CPS} = 0.66$ (model ratio 59.0% vs CPS 59%; Fig. 17). Absolute-level checks prefer $\alpha \geq 0.8$ but are level-biased: the colonial run overstates the immigrant stock ($C_0(2013) = 19.2\%$ vs CPS first generation 12.9%). Decision: $|0.66 - 0.8| = 0.14$ is not material → **keep** $\alpha = 0.8$ **as central**, bracketed by the ratio fit (0.66) and the absolute-series preference (≥ 0.9).

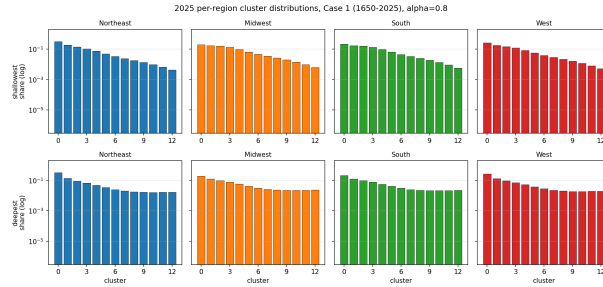


Figure 15: 2025 per-region cluster distributions (Case 1, $\alpha = 0.8$), log scale: shallowest (top), deepest (bottom). Midwest (11.3% immigration) deepest; Northeast (24.0%) shallowest.

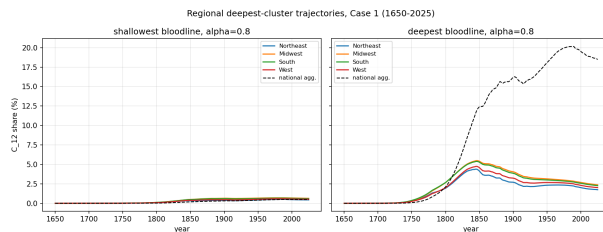


Figure 16: Regional C_{14} trajectories 1650–2025 ($\alpha = 0.8$), both rules plus national. Regions diverge after ~ 1800 , peak ~ 1980 , decline under immigration dilution.

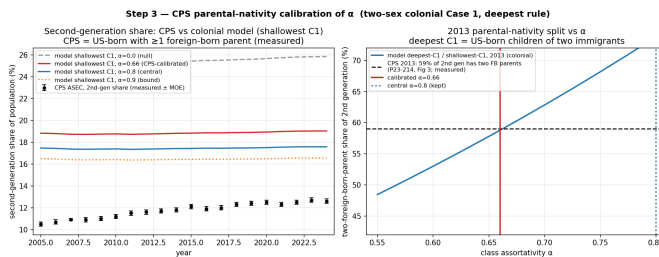


Figure 17: CPS calibration of α . Left: CPS ASEC second-generation share 2005–2024 (measured $\pm$ MOE) vs model shallowest C_1 . Right: 2013 two-parent share (deepest- C_1 /shallowest- C_1) vs α ; CPS 59% anchor (P23-214): calibrated $\alpha = 0.66$, central $\alpha = 0.8$ kept.

7.9 Religious dimension: composition and deepest clusters at $n = 14$

The religious extension (Sec. 2.2) was run 1650–2025 at $n = 14$ (240 ODEs) for both bloodline rules (shallowest and deepest) and $\alpha = 0.0, 0.8, 0.9$, with the group endogamies ρ_g of Table 1, the era immigrant compositions $\iota_g(t)$, and disaffiliation ramping 1970–1990 to the calibrated $\delta_{\max} = 8 \times 10^{-3} \text{ yr}^{-1}$. Totals are 258.95 M in 2025 in every run (religion is compositional: disaffiliation is internal), and the exact balance $dN/dt = I + B - (\mu + \varepsilon)N$ holds to $\sim 10^{-15}$ with disaffiliation active.

2025 religious composition vs. Pew. Modeled 2025 shares (deepest rule, $\alpha = 0.8$; identical across both rules and α to the printed precision, since group totals do not depend on the child-cluster rule) against the Pew 2023–24 Religious Landscape Study adult shares [9]:

	pro	cat	mor	jew	mus	dhr	oth	una
Model 2025	43.3%	17.9%	0.2%	3.2%	2.2%	2.7%	2.6%	27.9%
Pew 2023–24	40%	19%	$\sim 1.6\%$	2%	1%	2%	$\sim 1\%$	29%

The unaffiliated share is the calibrated quantity (27.9% vs. 29% Pew); the rest is a sanity check, not a fit. Mormon is under-shot (0.2% vs. $\sim 1.6\%$): the model has no conversion *into* Mormonism and no fertility differential, while disaffiliation drains the group at the full $\delta(t)$ — a named limitation. Figure 18 shows the full 1650–2025 trajectories with the Pew anchors.

Deepest-cluster shares at $n = 14$. The 2025 C_{14} share of the total population, by rule and α :

Rule	$\alpha=0.0$ (null)	0.8 (central)	0.9 (bound)
Shallowest inheritance	0.00%	0.45%	1.47%
Deepest inheritance	68.80%	17.93%	9.34%

The ordering and the α -direction match the two-sex $n = 14$ runs (Sec. 6.1); the two families agree to within 0.5 pp everywhere, so the religious extension is a genuine apples-to-apples refinement.

Deepest clusters by religious group. Within-group C_{14} shares in 2025 ($\alpha = 0.8$; Fig. 19) run from 24.51% (Protestant) and 18.65% (unaffiliated) down to $\approx 1.6\%$ for post-1965-immigrant groups (Muslim, Hindu/Buddhist) under deepest inheritance; under shallowest, all groups sit below 0.7%.

Case-2 religious projection, 2025–2050. From the Case-1 2025 state ($n = 14$): 2050 total 279.7 M; unaffiliated 27.9% → **36.4%** (rule-independent, the one-way disaffiliation flow continued at the calibrated $8 \times 10^{-3}/\text{yr}$ — a flagged artifact, not a forecast; two-way switching is future work, Sec. 7). 2050 C_{14} at $\alpha = 0.8$: shallowest **0.42%**, deepest **16.25%**.

7.10 Stochastic colonial era: the deterministic trajectory is unbiased

τ -leaping ensembles ($N = 300$ trajectories, 1650–1800, all- C_0 initial condition) under both inheritance rules, testing the ODE for finite-population bias, deep-cluster coefficients of variation, and discrete establishment years. Is the ODE a fair description while the colonial population is small (50,368 in 1650)? This section writes down the jump process, the τ -leaping algorithm, and the

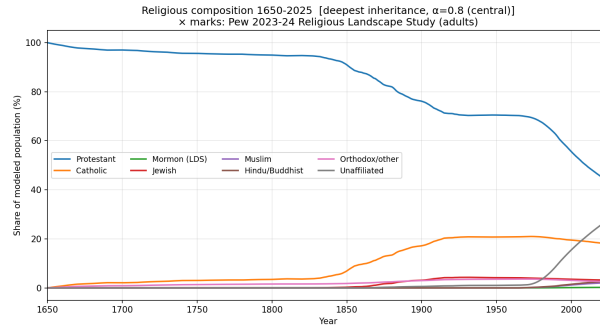


Figure 18: Religious composition 1650–2025 (deepest, $\alpha = 0.8$, $n = 14$). $\times$: Pew 2023–24 adult shares [9]. Unaffiliated (gray) is the calibrated quantity: disaffiliation ramps 1970–1990 to $\delta_{\max} = 8 \times 10^{-3}/\text{yr}$, landing at 27.9% vs. Pew’s 29%.

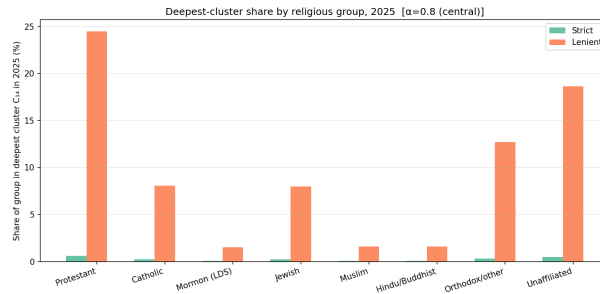


Figure 19: Deepest-cluster (C_{14}) share *within* each religious group in 2025, by rule ($\alpha = 0.8$, $n = 14$). Long-tenure groups run deep (Protestant 1.70%, unaffiliated 1.35%); post-1965-immigrant groups stay shallow ($\approx 0.17\%$).

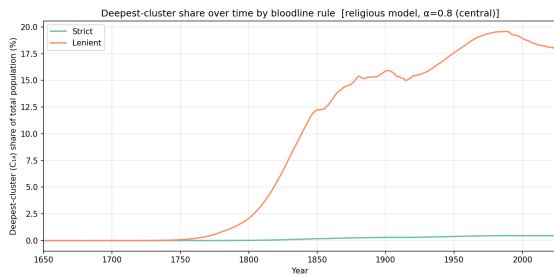


Figure 20: Deepest-cluster (C_{14}) share of the total population over time, by rule (religious model, $\alpha = 0.8$, $n = 14$).

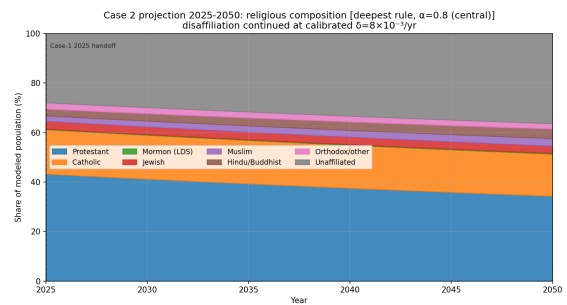


Figure 21: Case-2 religious composition 2025–2050 (deepest, $\alpha = 0.8$, $n = 14$). Unaffiliated 27.9%→36.4%; Protestant 43.3→34.3%; Muslim 2.2→3.0%; Hindu/Buddhist 2.7→3.8%.

simulation procedure in full. Is the ODE a fair description while the colonial population is small (50,368 in 1650)? This section writes down the jump process, the τ -leaping algorithm, and the simulation procedure in full.

The jump process. The state is the integer population vector $\mathbf{X}(t) \in \mathbb{N}^{2(n+1)}$ (people; block order [pat, mat]; 30 states at $n = 14$). Three reaction classes fire, with propensities (people/yr) and stoichiometry:

$$\emptyset \xrightarrow{a_{\text{imm},\alpha}} (0, \alpha), \quad a_{\text{imm},\alpha}(\mathbf{X}) = \sigma_\alpha(t) I(t), \quad \sigma_{\text{pat}} = \sigma(t), \quad \sigma_{\text{mat}} = 1 - \sigma(t) \quad (8)$$

$$(q, \text{pat}) + (r, \text{mat}) \xrightarrow{a_{qr,\alpha'}} (q, \text{pat}) + (r, \text{mat}) + (g(q, r), \alpha'), \quad a_{qr,\alpha'}(\mathbf{X}) = \begin{cases} s J_{qr}(\mathbf{X}), & \alpha' = \text{pat} \\ (1-s) J_{qr}(\mathbf{X}), & \alpha' = \text{mat} \end{cases} \quad (9)$$

$$(d, \alpha) \xrightarrow{a_{d,\alpha}} \emptyset, \quad a_{d,\alpha}(\mathbf{X}) = (\mu_\alpha(t) + \varepsilon) X_{d,\alpha}. \quad (10)$$

Parents are catalytic: they appear on both sides of the birth reaction and are not consumed. The mating flux J_{qr} is exactly the ODE kernel of Sec. 4,

$$J_{qr}(\mathbf{c}) = B(\mathbf{c}) [\alpha \delta_{qr} m_q^{\text{pat}} + (1-\alpha) m_q^{\text{pat}} m_r^{\text{mat}}], \quad B = \min(W^{\text{pat}}, W^{\text{mat}}), \quad W^{\text{pat}} = \sum_d \beta_{d,\text{pat}} c_{d,\text{pat}}, \quad m_q^{\text{pat}} = \frac{\beta}{\alpha} \quad (11)$$

evaluated here on the integer counts $\mathbf{X}$; the child-sex split $s : (1-s)$ in (9) is the exact Poisson thinning of the J_{qr} event stream.

Mean-field limit. The family is density-dependent in the sense of Kurtz [39]: immigration (8) and death (10) are linear in $\mathbf{X}$, and the mating flux (11) is homogeneous of degree 1 in $\mathbf{X}$ (the weights m are scale-free), so the scaled process $\bar{\mathbf{X}}^{(\lambda)} = \mathbf{X}^{(\lambda)}/\lambda$ converges to the Case-1 ODE trajectory $\mathbf{c}(t)$ as $\lambda \rightarrow \infty$. A $\lambda = 1000$ population-scaled check ($N = 30$ trajectories, shallowest, $\alpha = 0.8$) reproduces the ODE with total max |rel. dev.| = 7.7×10^{-5} , confirming the Kurtz limit in the code.

τ -leaping algorithm. Propensities are frozen over each leap (explicit Euler on the jump process [36, 37]). Given state $\mathbf{X}$ at t and step τ :

1. **Births:** for every mating-pair type (q, r) ,

$$K_{qr,\text{pat}} \sim \text{Poisson}(s J_{qr}(\mathbf{X}) \tau), \quad K_{qr,\text{mat}} \sim \text{Poisson}((1-s) J_{qr}(\mathbf{X}) \tau), \quad (12)$$

the child-sex split by exact Poisson thinning.

2. **Deaths/emigration:** exact per-leap for the constant per-capita rate over the leap,

$$L_{d,\alpha} \sim \text{Binomial}(X_{d,\alpha}, 1 - e^{-(\mu_\alpha(t) + \varepsilon)\tau}), \quad (13)$$

bounded by $X_{d,\alpha}$, so no state can go negative.

3. **Immigration:**

$$I_\alpha \sim \text{Poisson}(\sigma_\alpha(t) I(t) \tau). \quad (14)$$

4. State update (birth inflow accumulated by child cluster $g = g(q, r)$):

$$\mathbf{X}(t+\tau) = \mathbf{X}(t) + \sum_{q,r} \sum_{\alpha' \in \{\text{pat}, \text{mat}\}} K_{qr,\alpha'} \mathbf{e}_{(g(q,r),\alpha')} + I_{\text{pat}} \mathbf{e}_{(0,\text{pat})} + I_{\text{mat}} \mathbf{e}_{(0,\text{mat})} - \sum_{d,\alpha} L_{d,\alpha} \mathbf{e}_{(d,\alpha)}, \quad (15)$$

with $\mathbf{e}_{(d,\alpha)}$ the unit vector of state (d, α) .

The step is adaptive,

$$\tau = \min(0.5, \text{LEAP_EPS}/r_{\max}(t)) \text{ yr}, \quad r_{\max}(t) = \max(\beta, \mu) + \varepsilon, \quad (16)$$

in the Cao–Gillespie–Petzold style [38]: because J is homogeneous degree-1 with scale-free shares, bounding the relative state change ($\text{LEAP_EPS} = 0.002$) bounds the relative propensity change. Convergence tests show the τ error *compounds* through the generation chain: $\text{LEAP_EPS} = 0.05$ puts the 1800 C_{14} share 15% below the ODE, 0.01 puts it 3.2% low, 0.002 puts it 0.6% low (totals off by $1.6 \times 10^{-3}/4 \times 10^{-4}/2 \times 10^{-5}$) — so a too-large τ would mimic a “stochastic deficit”; at production τ the residual bias sits below the noise floor ($\sim 7,950$ leaps/trajectory).

Simulation procedure. Ensembles: shallowest/deepest $\times \alpha \in \{0.8, 0.0\}$, $N = 300$ trajectories, 1650–1800 (~ 3.9 M by 1800), from the all- C_0 1650 initial condition (50,368 people, σ -split between the sexes) — the same founding bottleneck as the deterministic colonial run. Drivers and ICs are identical to that run (inputs.csv, $\sigma(t)$, sex-specific $\mu(t)$, $\varepsilon = 7.5 \times 10^{-4}/\text{yr}$, $s = 0.512$). The ensemble is vectorized over the trajectory axis (all N trajectories advanced in one leap loop), pseudo-random numbers from NumPy’s `default_rng` with fixed seeds, yearly snapshots recorded; from them: ensemble-mean vs ODE z -scores and relative deviations, per-cluster CVs at 1800, extinction fractions (share of trajectories with $C_d = 0$), median establishment years, and the $\text{CV} < 1\%$ crossover. Four findings:

- **(i) No bias.** Ensemble-mean total vs ODE at 1800: $z = -0.84$ (shallowest), $z = +1.40$ (deepest), $\max |\text{relative deviation}| \leq 6.1 \times 10^{-4}$ over the whole 1650–1800 trajectory (Fig. 24). The mean-field check ($\lambda = 1000$, $N = 30$) reproduces the ODE total to 7.7×10^{-5} .
- **(ii) Deep-cluster mass robust.** CV of the 1800 cluster share at $\alpha = 0.8$: C_6 0.4–0.8%, C_{10} 0.9–2.2%, C_{14} 9.5% (shallowest) / 2.7% (deepest; Fig. 23) — the deep mass is thousands of people by 1800, not a noise artifact.
- **(iii) Establishment is discrete.** At $\alpha = 0.8$, 0/300 trajectories have empty C_{12} – C_{14} at 1800 (both rules); median establishment $C_6 \approx 1662$ –1668, $C_{10} \approx 1686$ –1701 (Fig. 25) — vs the ODE’s “all clusters > 0 from the first birth”.
- **(iv) Crossover.** $\text{CV}(\text{total})$ peaks 0.72% ~ 1710 (population ≈ 0.25 M) and *never reaches 1%*: total-population stochasticity is negligible throughout, bottleneck included (Fig. 22).

Verdict. No headline conclusion changes. The deterministic colonial trajectory is unbiased for totals and all cluster shares down to the noise floor; the ODE misleads only for clusters with expected occupancy $\lesssim 10$ people — far below any reported share.

7.11 Independent validation: mixed and informative, not confirmatory

Out-of-sample checks of 2025 class shares against Census Bureau data. All modeled numbers are *predictions* read from the archived regional colonial run ($\alpha = 0.8$, shallowest + deepest); observed data are *measured* 2023 ACS 1-year survey estimates (table-based summary files [40]). Designs were pre-specified; no new model runs were made.

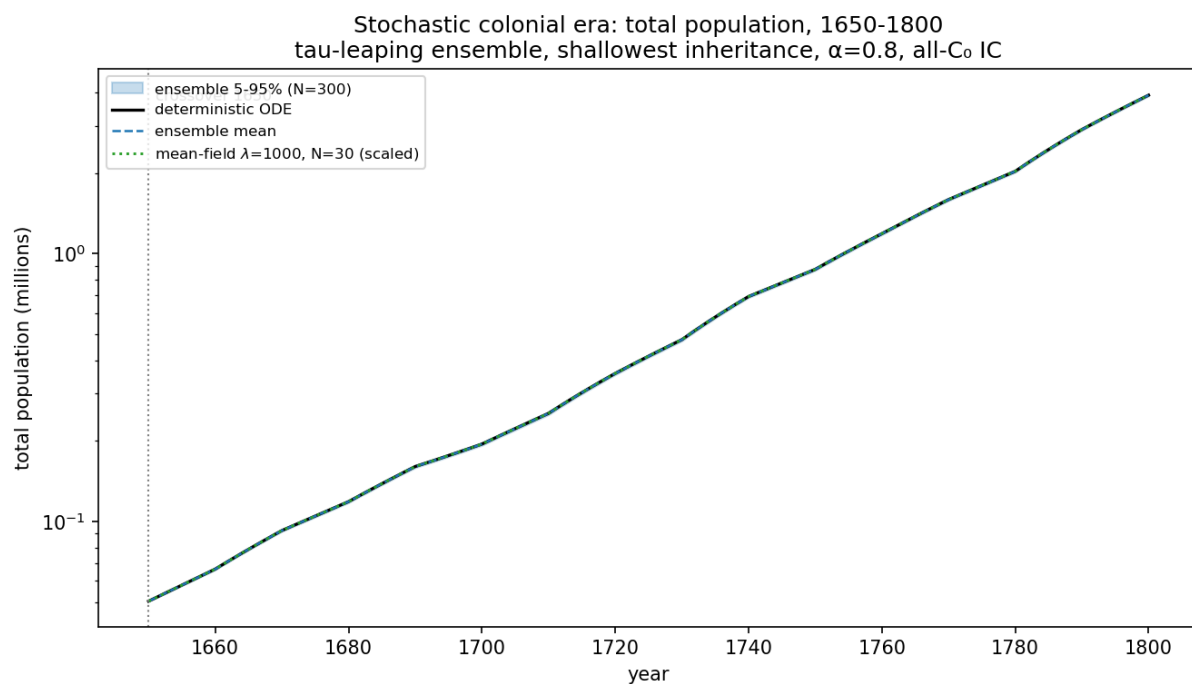


Figure 22: Ensemble fan 1650–1800, shallowest, $\alpha = 0.8$: 5–95% band (razor-thin) + CV(total) vs time; the 1% threshold is never reached (peak 0.72% at 1710, pop. ≈ 0.25 M).

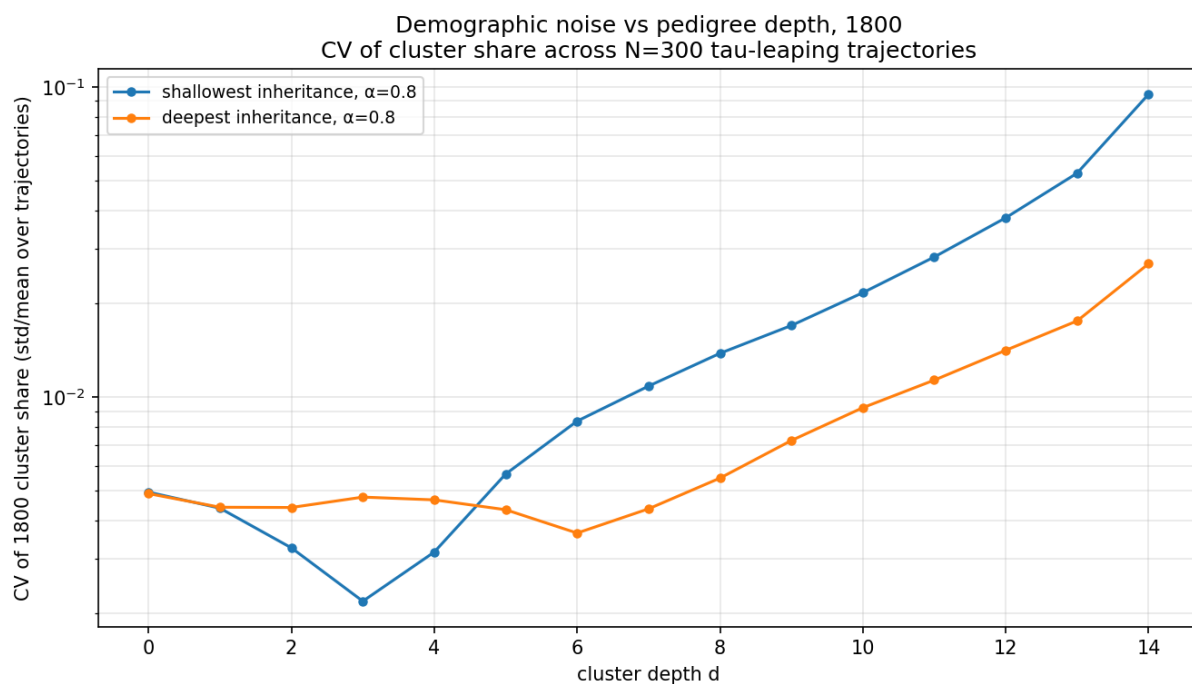


Figure 23: CV of the 1800 cluster share vs depth d , $\alpha = 0.8$ (both rules): C_6 0.4–0.8%, C_{10} 0.9–2.2%, C_{14} 9.5% (shallowest) / 2.7% (deepest) — deep-cluster mass is robust, not noise-dominated.

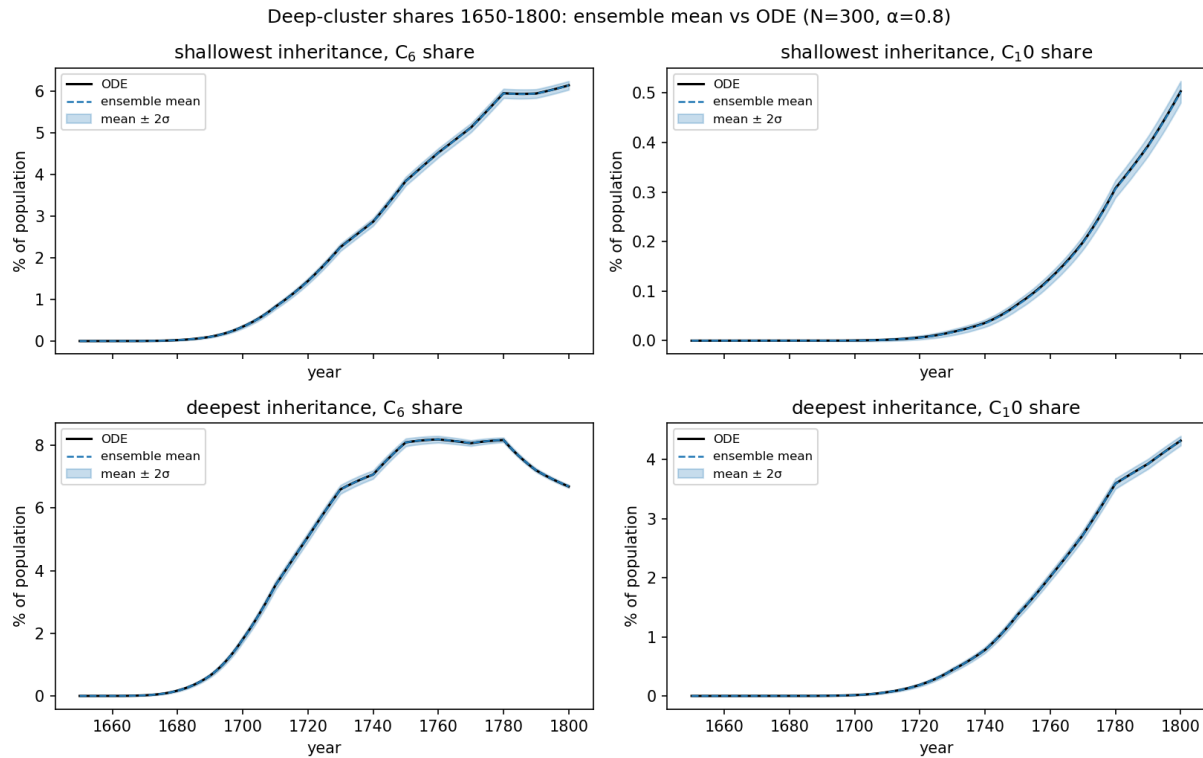


Figure 24: C_6/C_{10} share trajectories 1650–1800, $\alpha = 0.8$: ensemble mean $\pm 2\sigma$ vs ODE (both rules) — no bias ($|z| \leq 1.40$, max dev 6.1×10^{-4}).

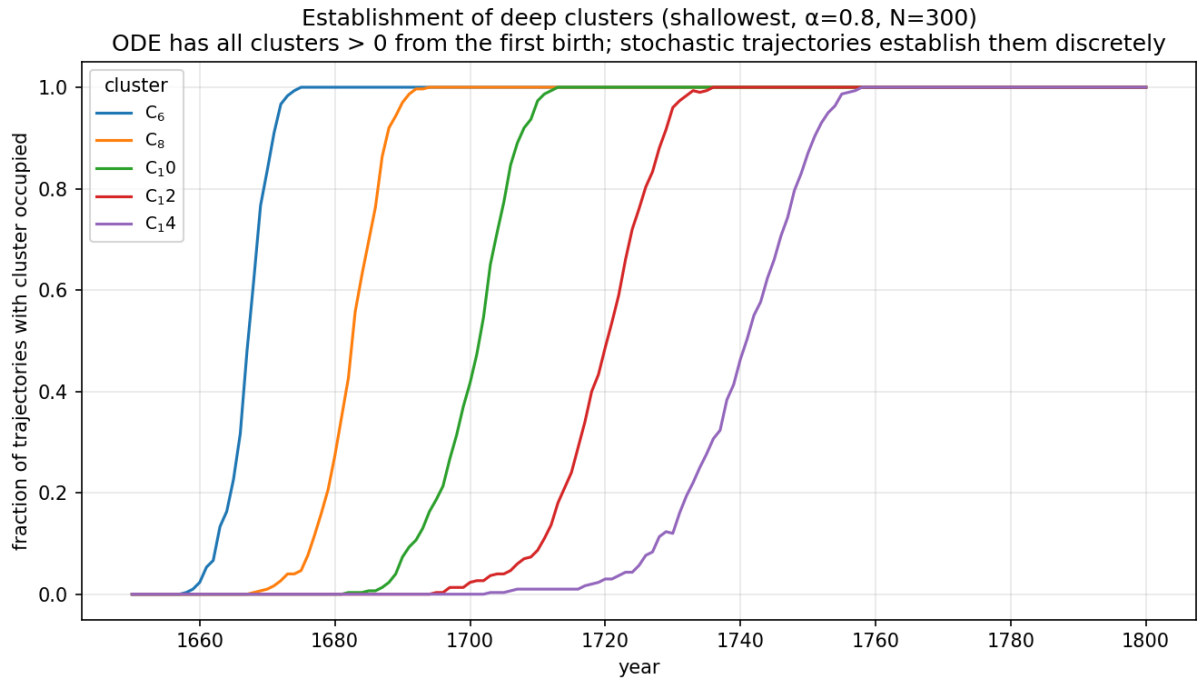


Figure 25: Fraction of trajectories with C_d occupied vs year (shallowest, $\alpha = 0.8$): establishment is discrete — vs the ODE’s “all > 0 from the first birth”.

(a) Regional “American”-ancestry check (primary, pre-specified). Observed: % reporting “American” ancestry by Census region (ACS B04006, measured) — a documented old-stock proxy, concentrated in the South. Modeled: regional 2025 C_{14} share, $\alpha = 0.8$, compared on rank order (Spearman ρ) and level. Result: **rank flip at the top** — observed $S > MW > NE > W$ (7.36 / 4.95 / 4.09 / 3.43%) vs modeled $MW > S > W > NE$ (shallowest 0.526 / 0.494 / 0.430 / 0.364%; deepest 20.61 / 19.54 / 17.20 / 14.74%; $\rho = 0.60$, $n = 4$). The model’s ranking is immigration-dilution arithmetic (the Midwest receives the fewest immigrants per capita, so deep lineages are least diluted); the proxy’s ranking reflects settlement history and cultural identity the model does not represent. This is a **real structural gap**, stated plainly. Level: the model underpredicts the proxy under shallowest (US 0.47% vs 5.38%) and overpredicts under deepest (US 18.47% vs 5.38%) — “American” self-report is broader than 14 consecutive U.S.-born generations but narrower than the deepest rule’s ratchet; descriptively the proxy’s national level lands near modeled shallowest C_{10+} (3.15%), i.e. $\sim 10+$ generations (post-hoc level match, not a confirmed prediction).

(b) Surname check: a genuine-attempt negative. No defensible operationalization was found with the public 2010 surname list (151,671 surnames [41]). No check was performed; documented as a negative result, not an omission.

(c) Foreign-born consistency check. Modeled C_0 exceeds observed foreign-born % by +38% (W) to +153% (MW), US +68%. The gap decomposes into documented pieces: the colonial run’s 2025 total (258.9 M) is 24% below actual (~ 342 M) while the immigrant stock is data-driven; model immigration inputs include DHS unauthorized estimates that ACS undercounts. Rank order largely holds ($\rho = 0.80$).

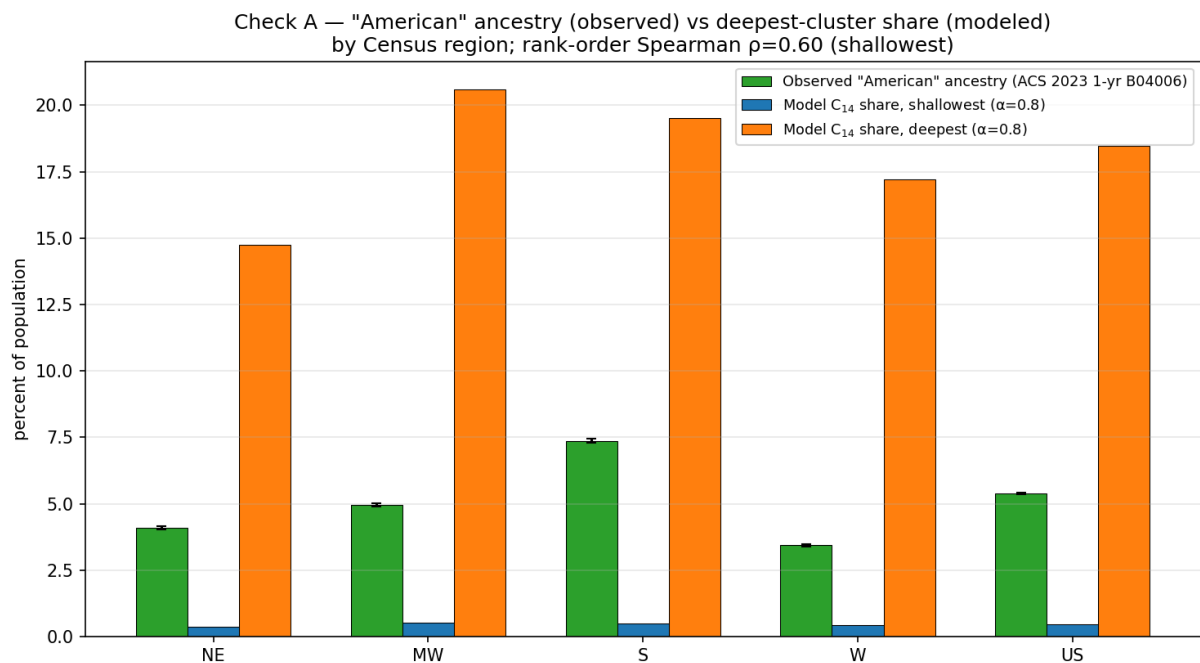


Figure 26: Check (a): observed “American”-ancestry % (2023 ACS, $\pm$ MOE) vs modeled 2025 C_{14} %, by region + US ($\alpha = 0.8$). Rank flip at the top: model peaks MW (immigration-dilution arithmetic), the proxy peaks S — a real structural gap ($\rho = 0.60$).

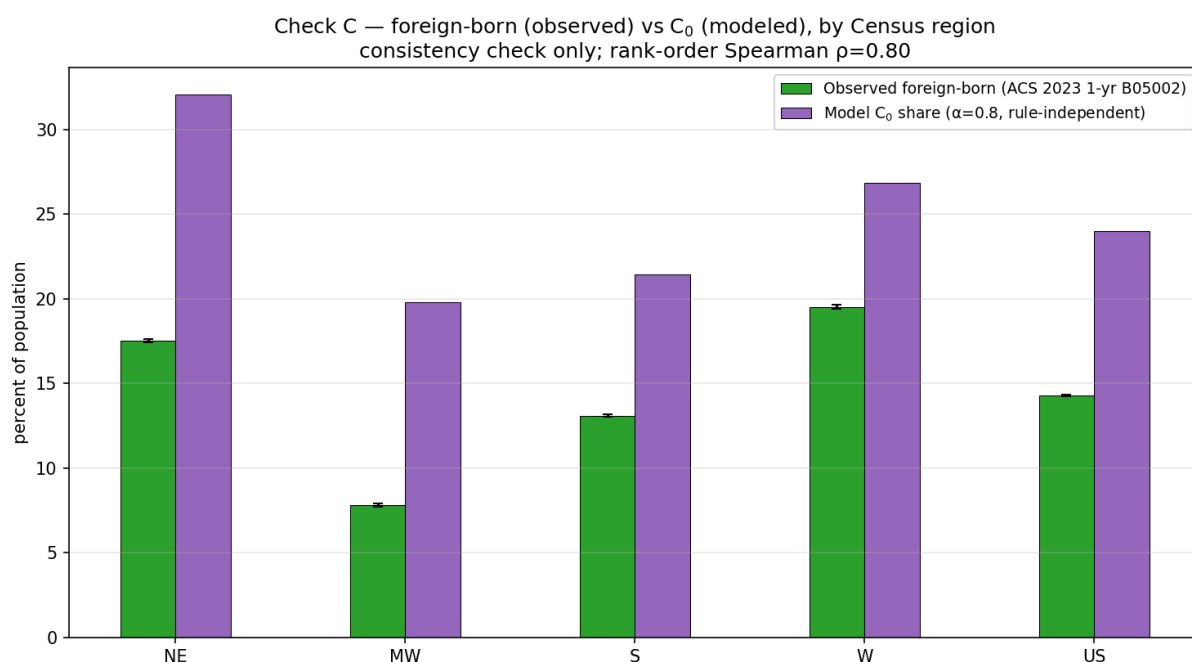


Figure 27: Check (c), consistency only: observed foreign-born % (2023 ACS) vs modeled 2025 C_0 %, by region + US. Level +38–153% (decomposes into the known total bias + unauthorized accounting); rank $\rho = 0.80$.

8 Suggestions for more accurate models and data

Completed since the last revision: age structure (Secs. 3.4, 7), regional compartments (Secs. 3.5, 7), the CPS parental-nativity calibration of α (Sec. 7.8), time-varying assortativity $\alpha(t)$ (Sec. 6.3, sensitivity variant), sex-specific rates with the maternal-bottleneck finding (Secs. 2.1, 6.6), deeper cluster resolution $n = 14$ with the tail study (Sec. 6.7), the stochastic colonial era (Sec. 7.10), and independent validation (Sec. 7.11).

1. **Age $\times$ religion interaction.** The 540-state age model and the 240-ODE religious model are verified unbroken but not yet combined.
2. **Regional vital-rate differences.** Regional fertility/mortality differences and cluster-specific migration are unmodeled; regional totals are uncalibrated.
3. **Paternal age schedule.** The male-fertility schedule is still the female ASFR shape (*assumed*); a ~ 2 yr older paternal schedule moves 2025 C_n by 2–5 pp at $\alpha \geq 0.8$ (Sec. 6.4). Calibrate from NCHS fatherhood data.
4. **$\alpha(t)$ for the extended models.** The time-varying assortativity schedule is implemented only for the two-sex model; port it to the religious, age, and regional models.
5. **Re-affiliation in disaffiliation.** The one-way disaffiliation flow is an honest artifact (unaffiliated 27.9% $\rightarrow$ 36.4% by 2050); add re-affiliation (two-way religious switching).
6. **Settlement-history process for regional deep shares.** Step 8 found a real structural gap: regional deep-cluster shares are set by immigration-dilution arithmetic with no region-specific old-stock history or identity process — the model peaks in the Midwest while observed “American” ancestry peaks in the South (Sec. 7.11). Add a settlement-history mechanism (or an identity/reporting layer) before the regional deep ranking can be trusted.

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